

Input Observed x
and y Lists

*Desired Latex
Equation*

*User Set MCMC
Parameters*

Optional Parameters

LaTeX
Equation Parser

Continuous
Function

Least
Squares
Fit

Initial
Coefficients

Vertical
Residuals

Initial Intrinsic
Scatter

Function Not
Continuous

Return
Error

False

$n_{\text{step}} = N_{\text{desired}}$

True

Posterior

Prior

Rejected

$n_{\text{step}} = n_{\text{step}} + 1$

Accepted

Calculate
Determinant &
Mahalanobis
Distance

Calculate
Gauss-Legendre
Quadrature
Moments

Sum to
Get Total
Likelihood

momus

Perform a Markov-Chain Monte-Carlo simulation to obtain the posterior over the desired coefficients of an inputted LaTeX equation while accounting for intrinsic scatter.

